

Day 5: Case Study Practical

Task 1 : A user reports their laptop won't power on even when plugged in. List 5 troubleshooting steps you would take to diagnose the cause, and for each step, explain what you are checking for.

Step	Troubleshooting Action	What You Are Checking For
1. Check the power outlet and charger	Plug another device into the same outlet and inspect the laptop charger, cable, and adapter for damage.	Checks whether the wall outlet is working and whether the charger is physically damaged or disconnected.
2. Check the charging indicator	Connect the charger to the laptop and observe the charging/power LED.	A missing LED may indicate a faulty charger, damaged charging port, or a power-input problem.
3. Test the charger with a multimeter	Set the multimeter to DC voltage and measure the charger's output, comparing it with the voltage printed on the adapter label.	Checks whether the charger is actually supplying the required voltage. A significantly incorrect or zero reading indicates a faulty adapter.
4. Perform a power reset	Disconnect the charger, remove the battery if removable, then hold the power button for about 15–30 seconds . Reconnect the charger and try powering on.	Clears residual electrical charge and can resolve temporary power-management or embedded-controller issues.
5. Inspect the laptop's power/battery connection	Check the charging port for looseness or physical damage and, if accessible, inspect the battery connection. Try a known-good compatible charger if available.	Determines whether the problem is caused by a damaged charging port, loose battery connection, faulty battery, or other internal power-circuit problem.

Task 2 : You receive a helpdesk ticket: 'My monitor is showing a blank screen, but the CPU seems to be running.' Write a step-by-step troubleshooting flow you would follow to identify and resolve the issue.

Hint: Consider checking both hardware connections and basic display settings.

Step-by-Step Troubleshooting

1. Check the monitor power

- Make sure the monitor is switched ON.
- Check whether the power LED is illuminated.
- Verify that the monitor's power cable is securely connected.
- **Checking for:** A monitor power problem or faulty power connection.

2. Check the display cable

- Check the HDMI/DisplayPort/VGA cable at both the monitor and computer ends.
- Disconnect and reconnect the cable firmly.
- If possible, try another known-good cable.
- **Checking for:** A loose, damaged, or faulty display cable.

3. Check the correct input/source

- Open the monitor's input/source menu.
- Select the port actually being used, such as **HDMI 1, HDMI 2, or DisplayPort**.
- **Checking for:** The monitor being set to the wrong input source.

4. Check the correct video output on the PC

- If the computer has a **dedicated graphics card**, make sure the display cable is connected to the graphics card's video port rather than the motherboard's port.
- **Checking for:** The display being connected to an inactive or incorrect video output.

5. Test with another monitor or TV

- Connect the PC to another known-working display using a compatible cable.

- **Checking for:** Whether the problem is with the monitor/cable or with the computer's graphics output.

6. Check keyboard/display shortcuts

- On Windows, press **Windows + P** and select **PC screen only** or **Duplicate**.
- If using a laptop, try the appropriate **Fn + display-key** combination.
- **Checking for:** Incorrect display mode or output being sent to another screen.

7. Check brightness and monitor settings

- Use the monitor's buttons/menu to verify that brightness is not set extremely low.
- If necessary, reset the monitor to its factory/default settings.
- **Checking for:** Basic monitor settings causing the blank display.

8. Restart and check for POST/display output

- Restart the computer and observe whether the manufacturer's logo or BIOS screen appears.
- If there is no display even during startup, check RAM/GPU connections and diagnostic beep codes or motherboard LEDs.
- **Checking for:** A hardware problem rather than a Windows/display-setting problem.

Resolution Flow

Monitor power → Display cable → Input source → Correct GPU port → Test another display → Windows display settings → Restart/POST → Hardware diagnosis

Conclusion

Start with simple external checks such as **power, cables, and input selection**. If these are working, check **Windows display settings**. If the screen remains blank even during BIOS/POST, investigate **RAM, GPU, motherboard, or other hardware components**.

Task 3 : A user complains that their wireless mouse is unresponsive, but the keyboard works fine. Describe how you would isolate whether the issue is with the mouse, the USB port, or the system settings.

Step	Action	What I am Checking
1. Check the mouse power	Turn the mouse OFF and ON. Replace/recharge the battery and check whether the mouse power/LED indicator works.	Checks whether the mouse has power or a dead battery.
2. Check the wireless receiver	Remove the USB receiver and reconnect it firmly. If possible, try another USB port.	Checks whether the receiver is loose or the original USB port has a problem.
3. Test another USB port	Connect the mouse receiver to a different known-working USB port.	If it works there, the original USB port may be faulty or disabled.
4. Test the mouse on another computer	Connect the wireless receiver to another working computer and test the mouse.	If it fails on another computer too, the mouse/receiver is likely faulty. If it works, the original system is likely the issue.
5. Check Device Manager	Open Device Manager → Mice and other pointing devices and Universal Serial Bus controllers . Look for warning/error symbols.	Checks whether Windows detects the mouse/receiver and whether there is a driver or USB-device problem.
6. Check mouse settings	Open Settings → Bluetooth & devices → Mouse and verify that the mouse is detected and settings are normal.	Checks whether the mouse is disabled or affected by incorrect system settings.
7. Reinstall/update the driver	In Device Manager, uninstall the affected mouse/receiver device and restart the PC so Windows can reinstall the driver.	Checks whether a corrupted or malfunctioning driver is causing the problem.

8. Final isolation test	Test a known-good wireless mouse on the same PC and USB port.	If the known-good mouse works, the original mouse/receiver is faulty. If it also fails, investigate the USB port, drivers, or system configuration.
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Diagnosis

- **Mouse fails on every computer:** likely **faulty mouse/receiver or battery**.
- **Mouse works on another computer but not this PC:** likely **driver/system settings or USB issue**.
- **Mouse works in another USB port:** likely **problem with the original USB port**.
- **Another mouse also fails on the same PC/port:** likely **USB port, driver, or system configuration issue**.

Conclusion: I would first check the mouse's power and receiver, then use **different USB ports and another computer** to quickly separate a mouse hardware problem from a USB-port or Windows configuration problem.

Task 4 : Imagine you are working on a helpdesk team for a gaming cafe. One system is overheating and shutting down during use. List 3 possible hardware causes and describe how you would check each one.

Hint: Think about fans, dust, and thermal paste.

Hardware Cause	How I Would Check It	What I Am Looking For
1. Faulty or non-working cooling fan	Open the PC case and check whether the CPU and GPU fans are spinning properly while the system is running. Listen for unusual noise and check fan speed/temperature using a monitoring tool such as HWMonitor.	A stopped, slow, noisy, or damaged fan can prevent proper heat removal and cause the PC to shut down when temperatures become too high.
2. Dust buildup	Power off and unplug the PC. Inspect the CPU heatsink, GPU heatsink, case fans, and air vents for dust. Clean accumulated dust using appropriate compressed air while preventing the fans from spinning freely.	Heavy dust can block airflow and insulate the heatsink, causing CPU/GPU temperatures to rise during gaming.
3. Dried or improperly applied thermal paste	If temperatures remain high after checking fans and cleaning dust, inspect the CPU cooler installation and thermal paste. Remove the cooler if necessary, clean the old paste, apply a suitable amount of new thermal paste, and reinstall the cooler correctly.	Dried, insufficient, or poorly applied thermal paste can reduce heat transfer between the CPU and heatsink, causing overheating under gaming load.

Conclusion

I would first check the **fans**, then **dust and airflow**, and finally the **thermal paste/cooler contact**. After each correction, I would run the system under gaming load and monitor CPU/GPU temperatures to confirm that the overheating and shutdown problem has been resolved.